

Day 15 (16/02/26)

• sinusoidal freq and just freq - diff

• Continuous signal $\longrightarrow$ discrete



if reconstruction successful, no info lost

$\rightarrow$ Fourier series can be used

$\rightarrow$ not use lossy method

$\rightarrow$ method should exist even if sampling rate not

following Nyquist rate

$\hookrightarrow$ before 2008 - couldn't (1920-2008)

after 2008 - signals should have some

properties { not gonna be discussed }

• Reconstruction $\equiv$ interpolation

$\hookrightarrow$ this will also tell us why: $f_s > 2f_{\max}$

• Dirac ~~delta~~ - sudden disturbance

not observed in nature

Impulse signal

Sharp transient for small time

Fourier Transform of signals

• Rect $\longrightarrow$ Sin function

Triangle $\longrightarrow$ sinc² function

$\hookrightarrow$ What happens if limit? Try

$-a/2$ to $a/2$?

also when center shifted?



it

out

convolving in time

*

$\equiv$ multiplication in

freq

(for rect) $\rightarrow$ (only phase changes)

$\rightarrow$ Scaling up in time $\equiv$ Scaling down in freq



domain scaling i.e. $n=1$

new $n=2$

$\hookrightarrow$ time axis

• Signal - finite duration signal

↳ Can we create another signal which makes it periodic?

→ Yes

→ repeat the signal after some period P

→ for infinite time

→ we can get original finite duration signal by marking the periodic signal

NOTE: $S_p(n) \neq f(n)$ • both are diff signals

• What is the convolution of dirac with a signal?

⇒ the signal itself - 0-05 min

⇒ What is the use of dirac then?

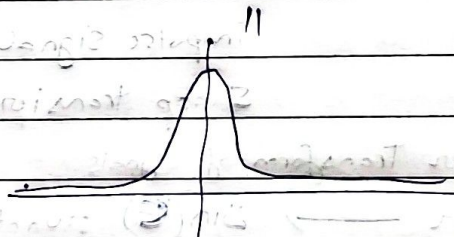
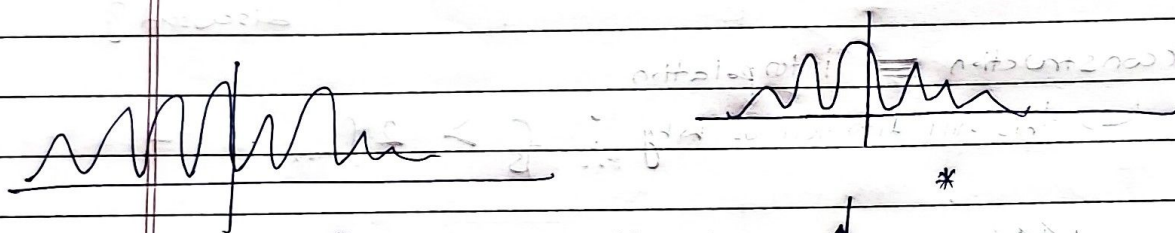
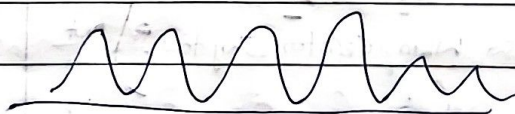


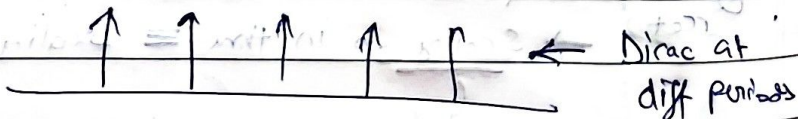
fig 1

fig 2

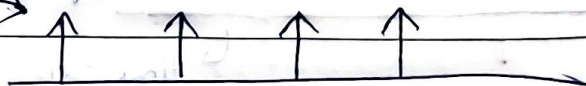
if dirac shifted



⇒



(Train of dirac)



• Train of dirac $\equiv$ Shah function



fourier transform of it
 $\equiv$ train of dirac

↓ why does it happen?



NOTE: fourier transform of dirac
is a single point / flat signal

if gaussian broad

$\Rightarrow$ variance high

$\Rightarrow$ infinite variance if flat

dirac - extreme form of

↓ discontinuity

difficult to obtain in

freq domain

as high in time

$\equiv$ low in freq

gaussian: broad in time

$\equiv$ narrow in freq

$$Ff = \Pi_P(Ff * \Pi_P)$$

fourier

transform

of signal f

periodizing it

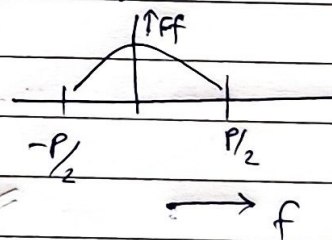
by multiplication

masking the periodic signal to get back the F.T of f

↳ by rect function

NOTE: Ff is also bounded (smthg) in freq

i.e.



Sampling

NOTE: sincp(t - tk) theorem

depend on $f(x)$

by the (P) period !!

as function was bounded.

It explains the nyquist sampling rate as well (check !!)

Q What does it

mean to involve?

(for communication)

NOTE: most systems work in discrete

like telephons etc but we

hear in continuous $\Rightarrow$ therefore

Sampling theorem req

Day 16 (20/02/26)

$$F^{-1} F f(u) = F^{-1} (\pi_p (F f * \pi_p))(t)$$

↑

for this equality to hold?

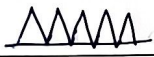
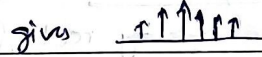
→ no overlapping of signals in f

→ bandlimitedness of f

• $f(t)$ gives values for non-sampled points as well after reconstruction from all sample points

• To periodise a signal, convolve it with a train of diracs

• NOTE: train of diracs in time and frequency same
except period in time is T ,
in freq it is $1/T$

• fourier series of  → gives 

• sampling $\Rightarrow$ multiplication! (with train of diracs)

NOTE: from a periodic

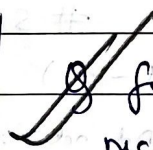
signal

↑
apply mask

↑

take fourier inverse

↳ gives $f(t)$

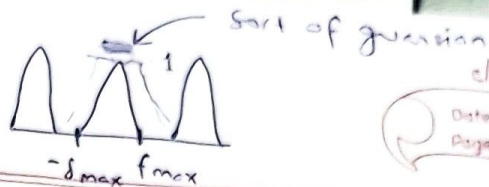


for convolution, is it necessary that one of them should be non periodic? (No)

NOTE: spectrums not corrupted: if 2 periodic signals

not overlap $\Rightarrow$

train of diracs should have time period $(\leq \frac{1}{T})$

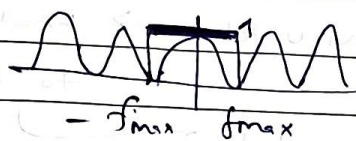


Date _____
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Disadvantage of oversampling — Samples increase; more data to store, more power also required

Advantage: redundancy — corruption easily handled
different mask we can use for getting the original signal

→ If at Nyquist rate $\Rightarrow$ we need rec function



Rec ~~func~~ in freq domain $\Rightarrow$ Sine in time

Why do we need in limited/finite interval?

→ not possible
→ so rec is not proper rec in freq domain $\therefore$ this is known as filter (the mask one)

kinds of filter to be discussed

$f(t)$ $\xrightarrow{\text{Sampling rate}}$ discrete form

Fourier transform

$\hat{f}(s) = F(f(t))$ $\xrightarrow{\text{how to discretise?}}$

Why do we do Fourier transform?

$\hookrightarrow$ Fast Fourier Transform

$\hookrightarrow$ What do we need to take care of?

(i) Integral $\rightarrow$ summation

(ii) infinite limit $\rightarrow$ finite

(iii) $f(t) \rightarrow$ discretize

finite in time

$\Rightarrow$ may be infinite in freq

• ex. rec

• if something decays smoothly in time

$\Rightarrow$ something happens in freq

→ Signals (in time)



train of dirac



Sampling (in time)



Apply mask to get original Spectrum

↳ but may be corrupted → what if we want to sample in freq?

(gives periodic ← take care that ←)

Signal train of ~~dirac~~^{time} doesn't overlap in time after sampling
(in time) (!)

- M — for indexing in freq
 N — for indexing in time

- Data if N } only relational
DFT matrix is $N \times N$

- DFT ensures minimum latency

↳ Implementation of it is FFT (Algo multiplication and addition req) — not more (!)

↳ previously it was N^2

Structure of DFT matrix

$N \times N$ DFT → $N/2$ DFT }
 $N/2$ DFT

→ further can be decomposed to 2×2

⇒ DFT should be power of 2 !!
matrix

Day 17 (23/12/26)

• Fourier Transform — complex terms

Topic: Can we remove noise from captured signals? — Denoising the signal?



how to use F.T.P.

⇒ F.T (captured signals) → DFT

High freq = (is noise)

how to do it in time domain?

→ Remove them → Take Inverse

⇒ apply gaussian filter

→ clean signal

⇒ moving average window — avg will take out the noise as it is moving very fast

Additive model

$$y[n] = x[n] + e[n]$$

← we should assume how noise coming into the model

$$y[n] = x[n] \cdot e[n]$$

noise removal separate for both AS or TSP

If we use additive

Approach to remove noise

⇒ we get the signal but it may not be clean completely

⇒ Some multiplicative component will always be there

$$\hat{y}[n] = \frac{1}{2m+1} \sum_{k=-m}^m y[n-k]$$

let's say $x[n] = \text{constant}$

noise = k and constant = 1

this helps in computing $e[n]$

we get →

→ (Cosine and sine)

Topic 2: • multiplying DFT matrix and data vector
≡ correlation computation of signal and
Sine waves of different freq

↑
this is what FT and PS used to do
here in discrete form we doing

• DFT only give indices to know frequency rate
← well should multiply by sampling rate

Topic 3: • DFT spectrum is symmetric around origin
↳ that's why focus on interval: $(n/2, 0)$
as symmetric - for real parts
[] not for complex

Topic 4: • After what frequency does the DFT decay?
↳ each spectrum will have it different

DFT in 2D

↳ projection of vectors in sine and cosine

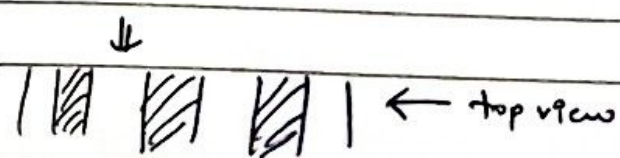
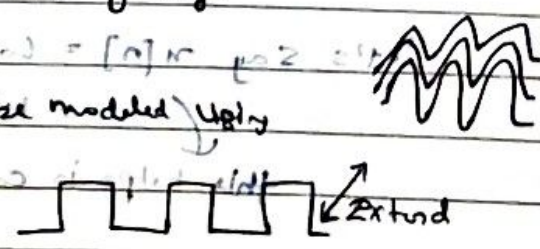
Q What is sine in 1D? ⇒ SHM (pendulum)
↓
tune ← since we can hear it

Example: [water waves]

Can we represent 2D signals in terms of sine and cosine?
→ 2D sine waves called gratings

NOTE: patterns can be modeled using

[functions]



• 2D FT

↳ $f(x, y) =$ summation of different freq signals in 2D

• 2D DFT

What is the use of it?

⇒ $\alpha, \beta, \gamma \dots$ tho α only need to be stored

⇒ jpeg compression works this way

Day 18 (24/02/26)

Low Pass filter

$$y[n] = x[n] + e[n]$$

$$= \alpha_1 \sin 2\pi f_1 n T_s + \alpha_2 \sin 2\pi f_2 n T_s$$

Zero out the signals which have freq > threshold

High Pass Filter -

ex. parabolic pattern - sine wave imposed on it

- Energy at 0 freq $\equiv$ Constant (from avg value)
⇒ high magnitude
- put a high pass filter to remove the parabolic pattern

$$y[n] = x[n] + e[n]$$

$$\Rightarrow = \left[a_0 + a_1 \left(\frac{T_s n}{T} \right) + a_2 \left(\frac{T_s n}{T} \right)^2 \right] + \beta \sin 2\pi f n T_s$$

a_1, a_2 due to sampling

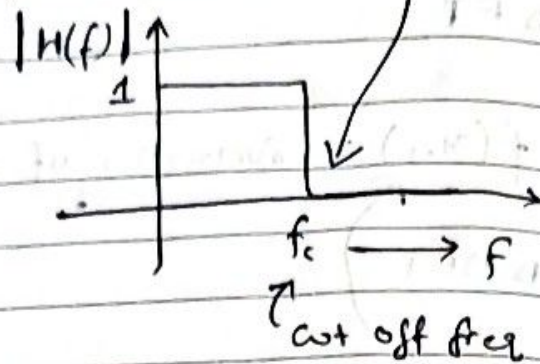
• Where to put the cut-off? - examined the spectrum

Filter Design

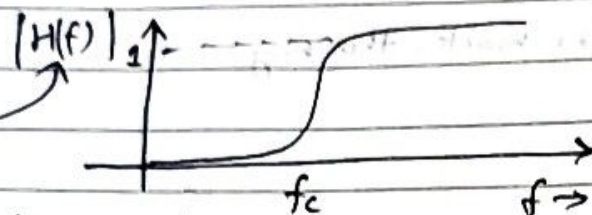
Freq response of the filter

Low pass filter

↳ in freq domain
it is applied

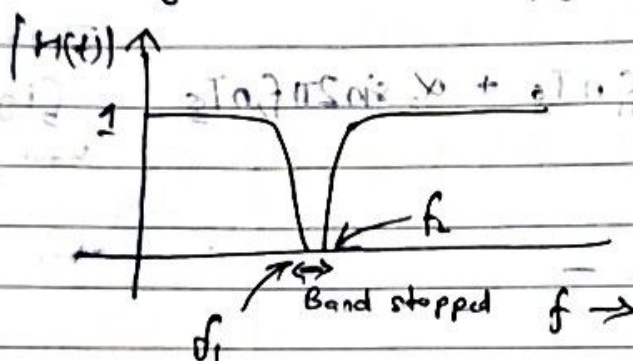


High pass filter



$H(f)$ is complex
but magnitude
only
plotted
(it has
phase as well)

Notch filter — Band stop filter

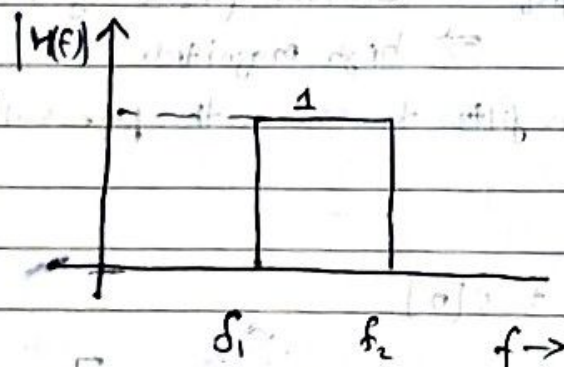


NOTE! if $f_1 = f_2$ here band is very small

⇒ Notch (pf)

⇒ ~~band~~ here no
signal is passed

Band pass filter



$$H(f) = \left[\frac{1}{2} \left(\frac{f - f_1}{f_2 - f_1} \right) + \frac{1}{2} \right] \left[\frac{1}{2} \left(\frac{f_2 - f}{f_2 - f_1} \right) + \frac{1}{2} \right]$$

Butterworth at cutoff

Butterworth at cutoff

You have 2 signals

And you make new signal

by take magnitude from one
and angle/phase from other

in FT and then apply FT^{-1}
(any Fourier transform can be written
as this)

$$x(t) \xrightarrow{[F]} X(f) = |X(f)| e^{j\angle X(f)}$$

$$y(t) \xrightarrow{[F]} Y(f) = |Y(f)| e^{j\angle Y(f)}$$

extract
magnitude from this

extract phase from
this

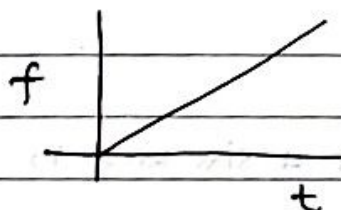
$$Z_1(f) \xleftarrow{FT^{-1}} X(f) e^{j\angle Y(f)}$$

- Resolution of signals improve if we
look for long time

What do we lose when we
go from one domain to another?

a) Time $\rightarrow$ freq

- from spectrum of $X(f)$ we can see //
- (i) here we know what freq, what magnitude we got
 - (ii) no info of time at which we observed the freq



$$x(t) = A_0 \sin \phi(t)$$

↑
phase

freq of this signal

$$x(t) = A_0 \sin 2\pi f t$$

$$\phi(t) = 2\pi f(t)$$

$$\phi(t) = 2\pi \int_0^t f(t) dt$$

how will $\phi(t)$ change? $\Rightarrow$ quadratic
as $f(t)$ is linear //

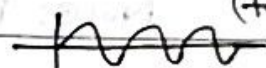
$\Rightarrow$ High freq $\Rightarrow$ Sampling rate not good

$\hookrightarrow$ Low Freq - able to sample properly

(check again)
• magnitude (IDROT)

• phase

• $\sin(\omega t + \phi)$, cosine
(terms)



• complex

• Why FT, why FT, why DFT,

Why DFT

(check again here) → Q how can we get the idea of what freq at what time the p. — (since not considered in DFT)

- don't look at whole signal
- look at small segment
- consider the time in that section //

↓ # Short time Fourier transform
here time is either the start or the middle element

1D signal → we can listen to it
2D signal → we can see the pattern

Spectrogram

NOTE: Put a window on the signal → Do FFT Over this

- bright colour
↳ more energy
- light colour
↳ low energy
- x-axis → time
- y-axis → freq
- high freq → low energy

- so we get freq for that duration
- plot that out

← observation

Q can we make a sine wave in Spectrogram?

- in time domain
↳ tuning fork
- freq should (inc, dec) x 10...
- in time domain, we need to do periodic sound

check harmonic T(as well)

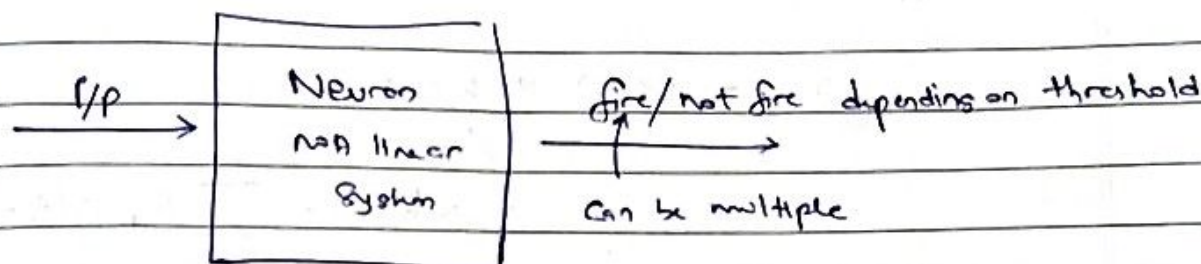
- ↳ which if observed in spectrogram can give us sine wave + harmonic signal comes up

Google Chromatic

Day 19 (10/03/26)

- Temporal Signal - Speech (anything happening in time)
- brain is plastic in nature - other part of brain starts functioning as defunct one
- Neuron (grey matter)
- brains release em waves (p)
- Dendrite - i/p
- Axon - o/p

- If current is high $\Rightarrow$ no. of Voltage firings is more } Rate encoding



Day 20 (13/03/26)

- Sound Signals

- $\rightarrow$ it is a time series : Seq of samples in time
- $\rightarrow$ Where is the info present in sound signal?

Even though reflection of sound happens
- why not able to hear? (maybe time difference)

Q What is microphone used for?
Recording or Speaking?

$$y(t) = \alpha_1 x(t) + \alpha_2 x(t-t_2) + \alpha_3 x(t-t_3) + \dots + \alpha_n (t-t_n)$$

attenuation

every source
will have

some time delay

$\Rightarrow$ Signal is same but reflections reach later //

(Sound modelling) $\rightarrow y(t) = \sum_{k=1}^n \alpha_k x(t - t_k)$

if $(t - t_k)$ is large $\Rightarrow$ it is echo
else it is reverberation/correlation

Can we express $y(t)$ as a convolution?

$\rightarrow$ Convolution of non-uniform Dirac with weights (determined by α_k)

$\rightarrow$ Dirac placed at the delay time

$$y(t) = x(t) * h(t)$$

$$h(t) = \sum_{k=0}^n \alpha_k \delta(t - t_k)$$

At $t=0$,

$$x(t) \alpha_k //$$

$$t_k = 0$$

$h(t) \leftarrow$ room's impulse response

($\Rightarrow$ it will change w/ room //

{ Convolution of signal
with an impulse at $t=0$
is same signal } *

$\Rightarrow$ if $h(t)$ is not known,
then get any impulse //

response)
(it's location specific
also)

Q How to understand the sound
will be perceived in a place?

$\rightarrow$ get the room's impulse response

$\rightarrow$

i.e. $y(t) = x(t) * h(t)$

(impulse)

(Same)



How to record sound

Day 20 (15/03/26)

$$y(t) = x(t) * h(t)$$

RIR

Can we obtain $x(t)$ from $y(t)$?
take in freq domain

$$Y(f) = X(f) \cdot H(f)$$

$$X(f) = Y(f) \cdot \left(\frac{1}{H(f)} \right)$$

take IFT

$$x(t) = y(t) * g(t) \quad \text{Note: } g(t) \neq \frac{1}{h(t)}$$

$g(t)$'s freq domain representation is $\frac{1}{H(f)}$

Data augmentation - with diff room impulse's response (not with noise)

$$y_1(t) = x(t) * h_1(t)$$

→ Record $x(t)$ in anechoic room

$$y_2(t) = x(t) * h_2(t)$$

(smth)

→ ~~the~~ convolve it with diff $h(t)$ available

$$y_n(t) = x(t) * h_n(t)$$

→ we get different sounds at diff locations

How do we produce sound?

Human ear - pinna helps for focusing sound, helps in

figuring out the elevation also as well location
provides elevation signature

• Hearing Impulse Response (HIR) - (of ears)

Similar to RIR

→ To understand how humans are ~~are~~ listening
like similar to how a sound will be perceived in a room //

Has to quantify audio (Time Series) with few features

focus on huge dimension — how to minimise?

What features

should we take

for sound?

long term features

↑

whole looking over the

period and

analyzing

(a) get top 3 frequency as features

• Convert to frequency → check the spectrum
↓

Pick top 3 freq

• time domain feature →

one kind of feature

• Compute energy (L2 Norm in DS)
• divide by N to make it independent of time

Q What feature in time domain can tell us about frequency?

Short term

another feature: • avg zero crossing rate

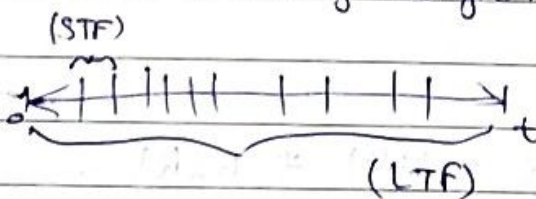
→ making histogram for the whole time

features

extract features from

short window //

Sensitive to duration



Period and Statistical evaluation etc

gives a matrix — per frame for ex. $(3 \times N)$

no. of features no. of frames

used when time not necessary!

Day 21 (17/03/26)

• HRIR - Head Related Impulse's Response

Q What is called a feature?



- it should define a property
- it should have distinction property
- it is informative
- it is task independent
- non stationary random process - pb distribution's statistical property changes ex. $0 - 0.5$ } mean and variance
 $0.5 - 1$ } changes significantly

• LTF - efficient;

disadvantage cannot assume T.S is stationary

• Spectrogram - Image analysis can be done!

• MFCC - male frequency cepstral coefficients

Q How do we distinguish between left and right sound?

→ Loudness in the audio source

→ time to reach ear (time diff)

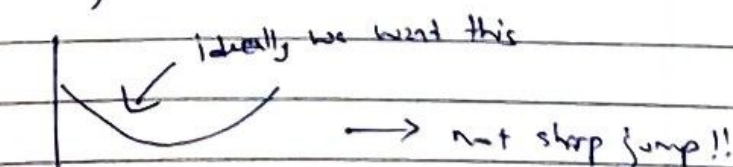
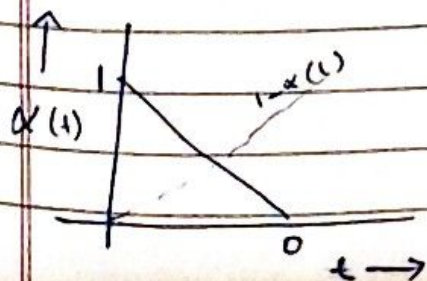
$x(t)$

(ILD) Interaural loudness difference

$$x_L(t) = \alpha(t) x(t)$$

$$x_R(t) = (1 - \alpha(t)) x(t)$$

(this is in loudness terms)



• ITD - interaural time difference

$$\begin{array}{l} \downarrow \\ \text{for left ear} \rightarrow x(t - \alpha(t)) \\ \text{for right ear} \rightarrow x(t - \alpha(t)) \end{array} \left. \vphantom{\begin{array}{l} \downarrow \\ \text{for left ear} \rightarrow x(t - \alpha(t)) \\ \text{for right ear} \rightarrow x(t - \alpha(t)) \end{array}} \right\} \text{original is } x(t)$$

• 2 channel audio signal - gives maximum perception,
Can be improved by perception of impulse of human ear

• Spectrogram - darker region means (low) energy //

• ITD - interaural time difference

for left ear $\rightarrow x(t - \alpha(t))$
for right ear $\rightarrow x(t - \alpha(t))$ } original is $x(t)$

• 2 channel audio signal - give movement perception,
Can be improved by perception of impulse of human ears

• Spectrogram - darker region means (low) energy //

Day 22 (20/03/21)

Perception of sound

- Loudness function of frequency
- Perception depends on the baseline + stimulus

$\rightarrow \approx \log(\text{Stimulus}) **$

for low intensity of light $\rightarrow$ we are sensitive
(almost linear)
(Luminance)

- How to find the threshold in which light detected?

(i) staircase method

$\rightarrow$ do sort of binary search

(start in extreme and then find the cut mark path)

example (ii)

in image if amplitude high $\rightarrow$ log applied //

$S \leftarrow$ perception/sensation

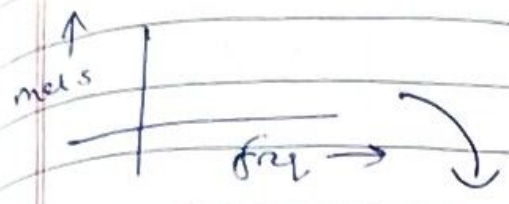
$S \rightarrow$ linear
 $S \rightarrow$ logarithmic
 $S \rightarrow$ exp

for each $1/p_2 \propto$ diffuses in this case //

- If gap increase is high bet 2 freq $\rightarrow$ we can have it as 2 diff sounds if $T \downarrow$

$m \rightarrow$ melody
(mels) $\rightarrow$ in graph

it represents the perceived freq v/s actual freq played



- Q How does cosine wave sound?
- Q How does GPT process the audio to text conversion? in it's voice chat feature

helps in retrieving MFCC $\uparrow$ mel

Similarly we can plot $\rightarrow$ loudness intensity v/s freq perceived

- perceived loudness depends on frequency (even if loudness of source kept same)

Q How do we know it is sine wave? $\downarrow$
giving periodic and oscillatory signal

Day 23 (23/03/26)

- Resolution at lower frequency is possible but not in higher frequency as frequency and perceived freq at lower freq is linearly dependent
- Reaction time for talker change detection - it's the time req to tell 2 people changed / speaker change
 $\rightarrow$ betⁿ male and female - it is less
but same type - it is more

(fMRI - Magnetic Resonance Imaging) similar to reverberation
↳ impulse ratio something

1. ((result))



(inspiration is supposed to be
(to 2 harmonics) ← 2me)

(We need to observe the lower part of sine wave / freq)

- * Note: harmonics don't come for every sound

2

— Vocal Cords Not

Voiced sounds $\xleftrightarrow[\text{stability difference}]{\text{harmonic}}$ (unvoiced sound)

opening $\Rightarrow$ energy
↓
or vibration

Voiced sounds — harmonic present!

⇒ but which amplitude observed is what is different for each alphabet //

Energy of each word/alphabet is diff

Vocal cords — provide harmonics

Who provides energy? — shape of the vocal tract //

What harmonics amplify or attenuate // it decides //

o Vocal cord $\xrightarrow{\text{harmonics}}$ Vocal tract $\rightarrow$ that's how words differentiated //

↑
amplify or
attenuate

o Whisper — doesn't have harmonics

→ is harmonic used for understanding the speech?

⇒ If harmonic is used $\rightarrow$ then whisper is still used differentiated ~~not~~ by brain!!